CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2020 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Four channels, each with 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 KHz between channels?

Solution: N = 4 channels of 100 KHz each.
Number of guard bands required between 4 channels = 4 - 1 = 3.
Each guard band = 10 KHz.
Total minimum bandwidth:
B = (4 \times 100) + (3 \times 10) = 400 + 30 = \mathbf{430 \text{ KHz}}

Q2 (3M) Discuss Point-to-Point Protocol (PPP).

PPP: A data link layer protocol used to establish a direct connection between two nodes over dial-up, serial cable, or broadband (PPPoE). Provides authentication via PAP/CHAP, framing, and dynamic IP configuration (IPCP).

Q3 (3M) Explain the functions of the Presentation Layer.

Functions: Data syntax translation, character code conversion, data compression (lossless/lossy), and data encryption/decryption (cryptography).

Q4 (3M) Differentiate between LAN, MAN, and WAN.

Comparison: LAN covers a room or building (high speed, private ownership). MAN covers a city (cable TV network). WAN covers a country or globe (telecom lines, satellite, Internet).

Q5 (3M) What is a Collision Domain vs Broadcast Domain?

Definitions: A Collision Domain is a network segment where simultaneous transmissions collide (divided by switches). A Broadcast Domain is a segment receiving all broadcast frames (divided by routers).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) How does DTE-DCE interface work? Discuss the essence of Modems.

DTE-DCE Interface: DTE (Data Terminal Equipment, e.g. computer/router) generates digital data. DCE (Data Circuit-terminating Equipment, e.g. modem) converts signals for transmission across the physical telephone link. Standards include RS-232 / EIA-232.

Q7 (7.5M) Explain Error Detection using CRC with generator polynomial $x^3 + x + 1$.

CRC Calculation: Complete modulo-2 polynomial division with bit representation `1011`.

Q8 (7.5M) Explain Distance Vector Routing Algorithm.

Algorithm: Periodic exchange of routing tables between adjacent neighbors based on Bellman-Ford algorithm.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain OSI Reference Model with detailed layer descriptions.**

7-Layer Model: Detailed functions, protocols, and data units (Bits, Frames, Packets, Segments, Data).

Q10 (15M) **Explain Sliding Window Protocols: Stop-and-Wait, Go-Back-N, and Selective Repeat.**

Flow Control: Efficiency equations, window buffering, and timeout retransmission mechanisms.

Q11 (15M) **Explain Transmission Media: UTP, STP, Coaxial, and Fiber Optic Cable.**

Transmission Media: Attenuation, propagation velocity, bandwidth, and EMI immunity analysis.

Q12 (15M) **Explain Congestion Control Algorithms: Leaky Bucket and Token Bucket.**

Traffic Shaping: Smooth constant-rate leakage vs bursty-tolerant token generation.

Q13 (15M) **Explain Switching Techniques: Circuit, Message, and Packet Switching.**

Switching: Detailed comparison of setup time, propagation delay, transmission delay, and queuing overhead.
